Linear Regression

The Quality Control department of a vineyard wants to develop a predictive model to estimate a wine's final sensory quality score based on various physicochemical factors such as acidity, residual sugar, sulfur dioxide, and alcohol content.

You are provided with the dataset WineQT.csv, which contains information about the chemical composition of wine samples and their corresponding evaluations.

Column	Description
fixed acidity	The concentration of non-volatile acids that do not evaporate readily
volatile acidity	The amount of acetic acid in wine, which can cause a vinegar taste at high levels
citric acid	Found in small quantities, adding freshness and flavor to wines
residual sugar	The amount of sugar remaining after the fermentation process stops
chlorides	The amount of salt present in the wine
free sulfur dioxide	The un-bound form of SO2, preventing microbial growth and oxidation
total sulfur dioxide	The total amount of both free and bound forms of SO2 in the wine
density	The density of the wine, varying by alcohol and sugar content
pH	A scale from 0 to 14 describing the wine's acidity level
sulphates	A wine additive that acts as an antimicrobial and antioxidant
alcohol	The percentage of alcohol content in the wine

Column	Description
Id	A unique numerical identifier for each wine sample observation
quality	The wine's sensory quality score (Target variable)

Your Task: Your task is to build a Simple Linear Regression model from scratch using NumPy to predict the quality of wine based on its alcohol content.

What to Do:

- **Load and explore the dataset.**
- **Preprocess the data — isolate the features and normalize numeric columns if necessary.**
- **Implement Linear Regression from scratch (do not use sklearn).**
- **Train your model to predict quality (target variable).**
- **Visualize the regression line against the data points or plot the residuals.**

Allowed Libraries:

- **Numpy**
- **Pandas**
- **Matplotlib**